O‘ZBEKISTON RESPUBLIKASI RAQAMLI TEXNOLOGIYALAR VAZIRLIGI

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OPERATSION TIZIMLAR 1-AMALIY ISH

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Tekshirdi: Bekimmetova Durdona

Guruh: SRM002-2

Toshkent – 2026

1-topshiriq. C tili asoslari, kompilyatsiya va pointerlar

Mavzu: C sintaksisi, "Hello World", kompilyatsiya bosqichlari, pointerlar (&, *, pointer arifmetikasi, funksiyaga pointer uzatish).

A qism (barcha talabalar uchun umumiy): "Hello World" dasturini 4 bosqichda alohida kompilyatsiya qiling (gcc -E, -S, -c, linking). Hosil bo'lgan .i, .s, .o fayllar va bajariluvchi fayl haqida hisobotda qisqacha izoh bering (har bir fayl nima ekanini va hajmini ko'rsating).

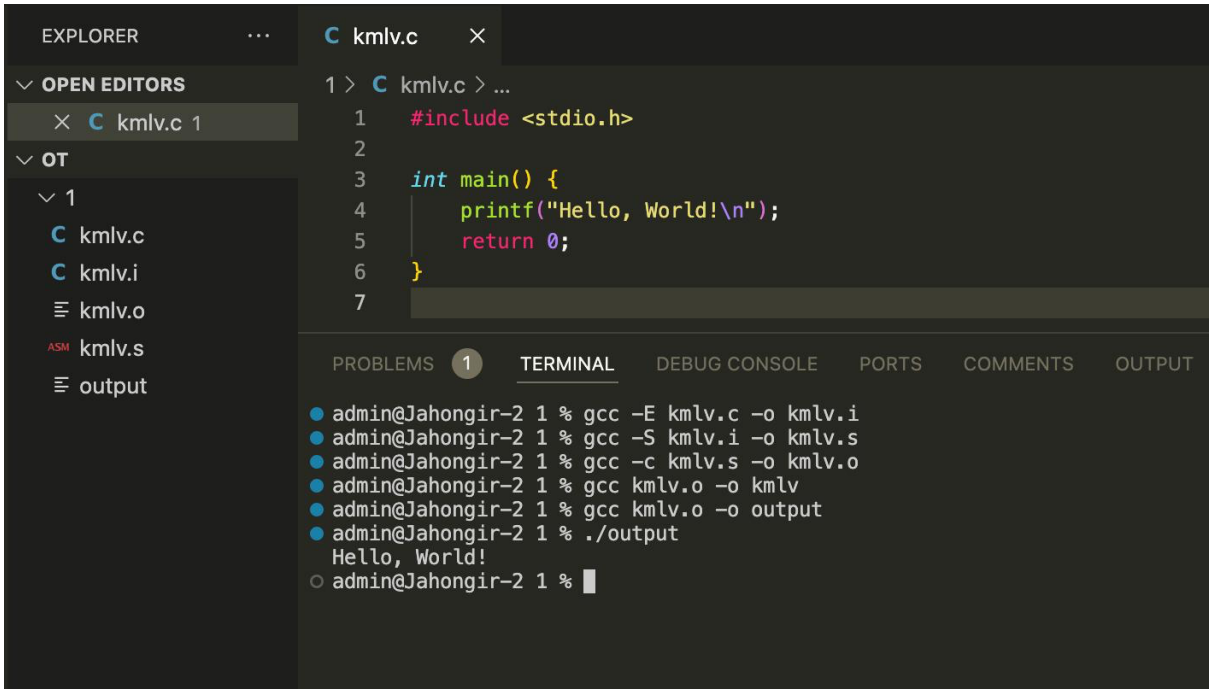
B qism (variant bo'yicha, ikkita masala): 1-masala C asoslariga, 2-masala shu masalani pointerlar yordamida kengaytirishga bag'ishlangan. Har bir masala kamida bitta alohida funksiya bilan yechiladi va kiritilgan ma'lumotlar tekshiriladi: masala uchun ma'nosiz qiymatlar (massiv o'lchami ≤ 0 yoki chegaradan katta, son o'rniga harf, mavjud bo'lmagan sana va h.k.) xabar bilan rad etiladi. Manfiy son yoki nol faqat masala shartida ma'nosiz bo'lsa xato hisoblanadi (masalan, kvadrat tenglama koeffitsiyenti yoki Selsiy harorati manfiy bo'lishi mumkin). 2-masalada:

- natijalar funksiyadan pointer orqali qaytariladi yoki massiv/satr elementlariga pointer arifmetikasi bilan murojaat qilinadi (bu 1-topshiriqda malloc talab qilinmaydi, dinamik xotira 2-topshiriqda);
- kamida bitta o'zgaruvchining manzili %p bilan chiqariladi va hisobotda izohlanadi.

Massivlar va satrlar: 1-topshiriqda malloc ishlatilmagani uchun massivlar statik e'lon qilinadi

(masalan, #define MAX 100; matritsalar uchun N, M ≤ 10). Kiritilgan o'lcham chegaradan oshsa,
xabar chiqariladi. Bo'shliqli satrlar fgets bilan o'qiladi (scanf("%s") bo'shliqda to'xtaydi) va
oxiridagi '\n' belgisi olib tashlanadi. <math.h> funksiyalari (pow, fmod, sqrt) ishlatilsa,
kompilyatsiyaga -lm qo'shiladi.

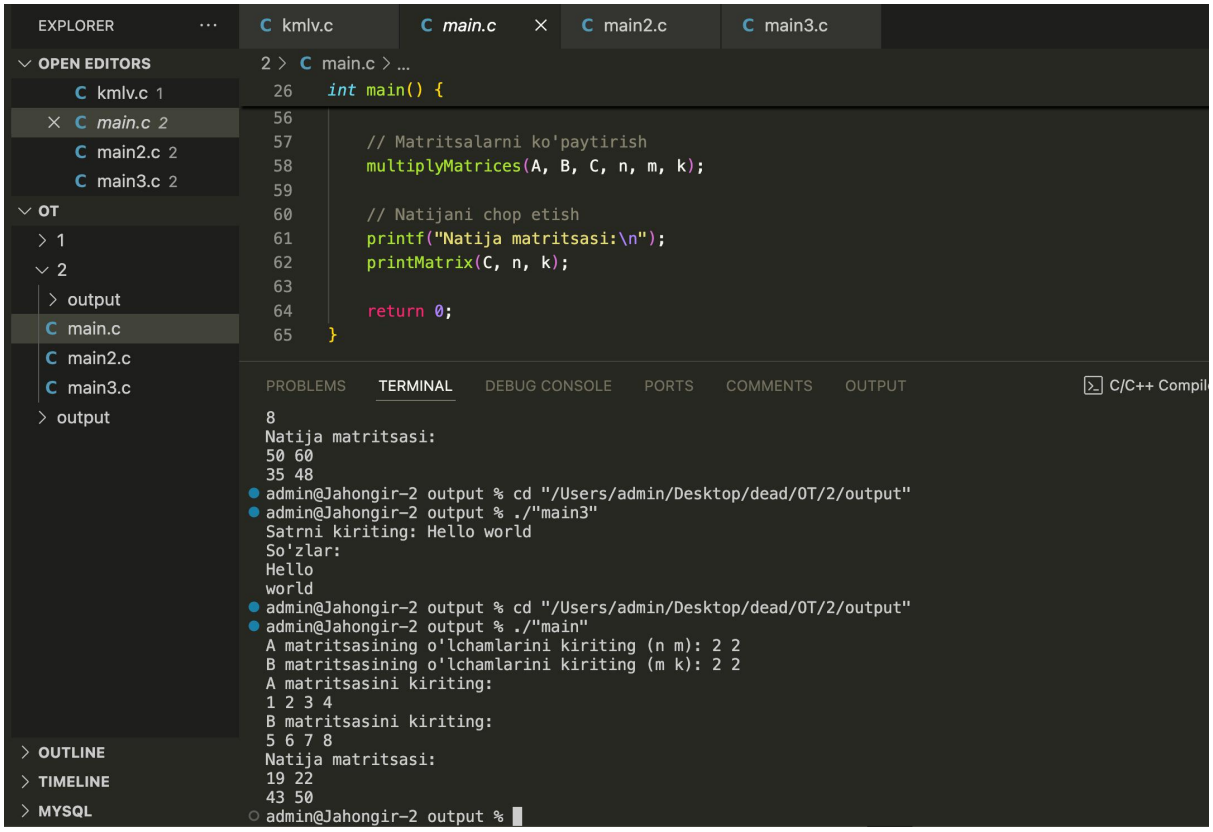
A.

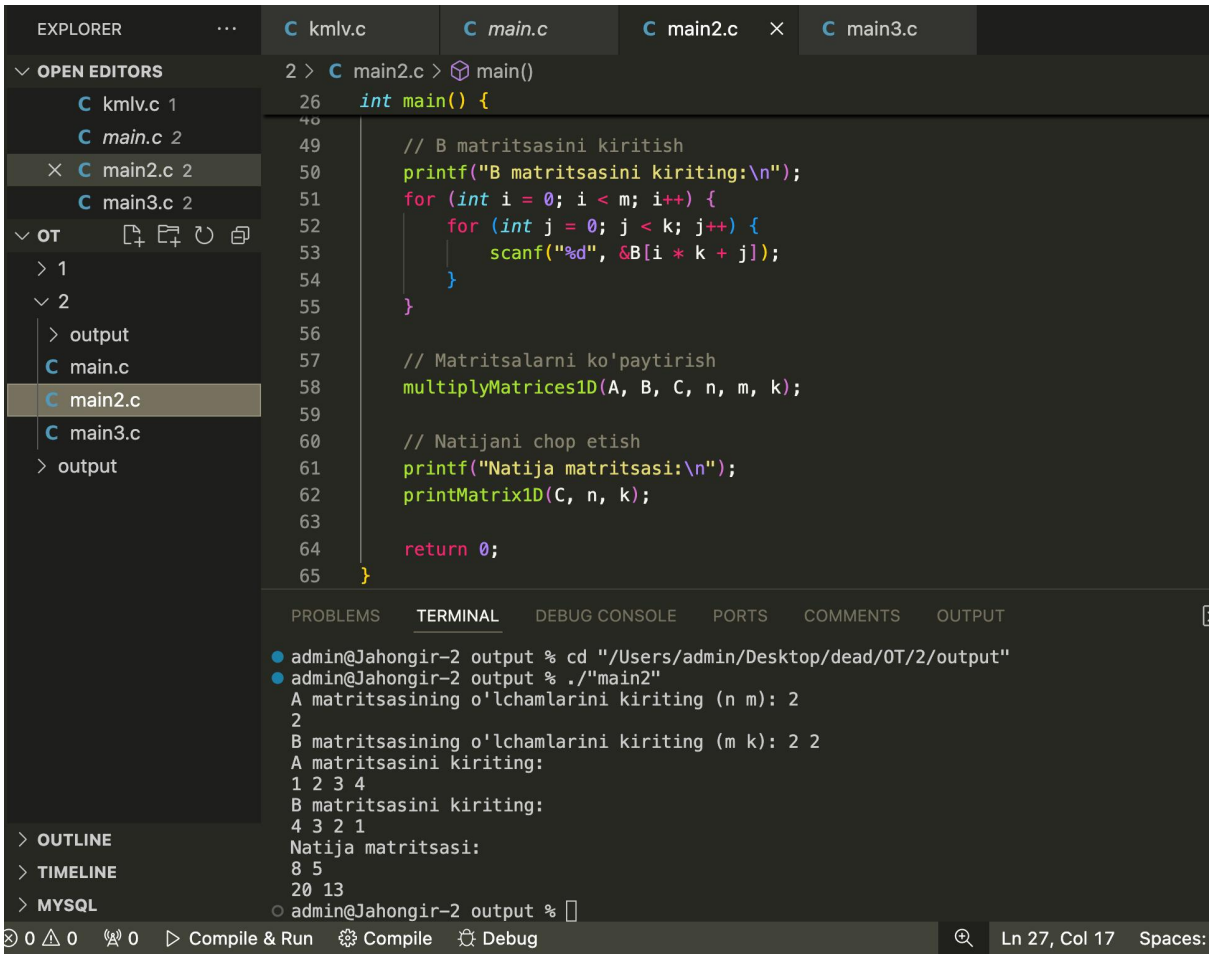


B.

2.1.

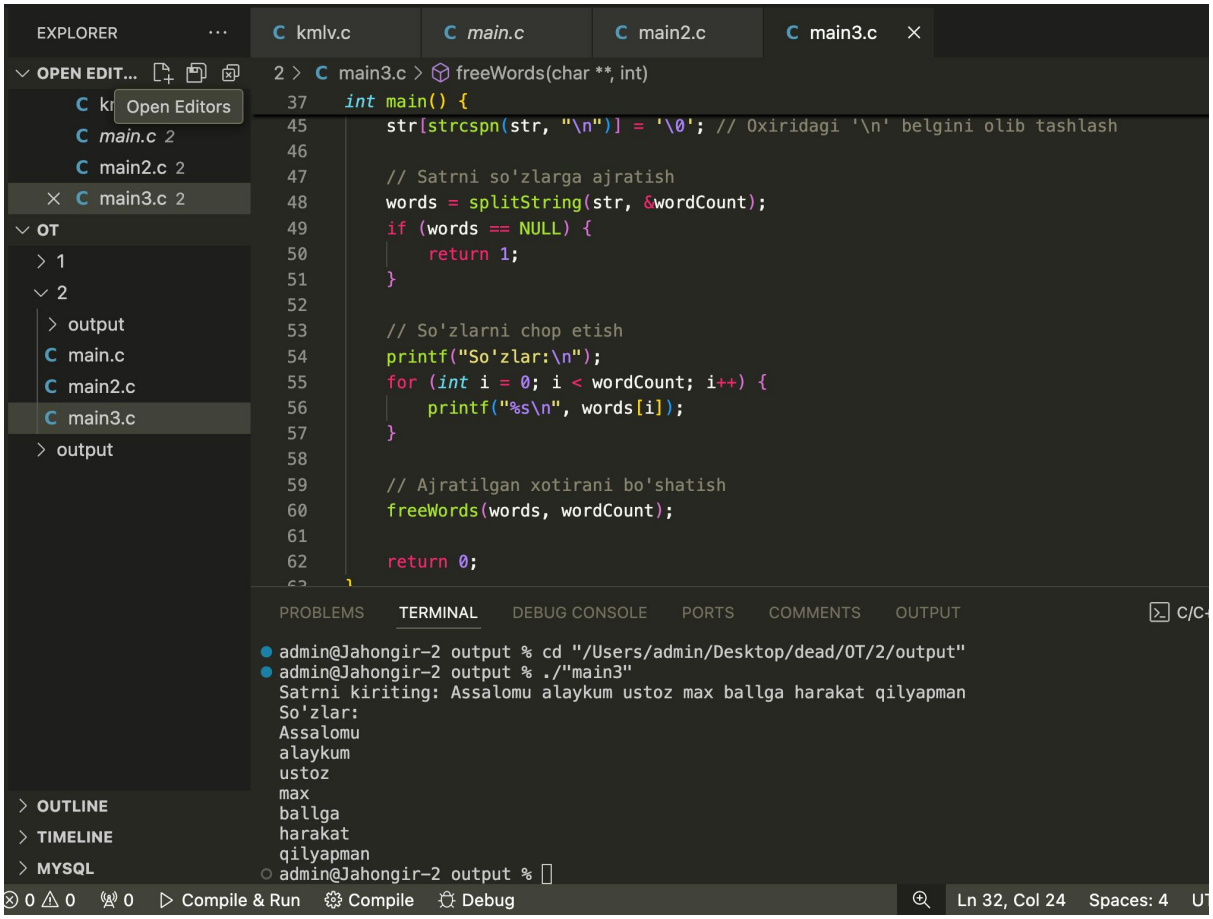
9	Ikki matritsani ko'paytiring (A: n×m, B: m×k); o'lchamlar mos kelmasa, xabar chiqaring.	Matritsalar bir o'lchamli massivda saqlansin (int a[MAX * MAX]). Ko'paytirish funksiyasi ularni const int *a ko'rinishida olsin va elementlarga a[i * m + j] orqali murojaat qilsin (m — A ning ustunlar soni).


EXPLORER
kmlv.c
main.c
main2.c
main3.c
OPEN EDITORS
kmlv.c 1
main.c 2
main2.c 2
main3.c 2
OT
1
2
output
main.c
main2.c
main3.c
output
OUTLINE
TIMELINE
MYSQL
2 > C main.c > ...
26 int main() {
56
57 // Matritsalarni ko'paytirish
58 multiplyMatrices(A, B, C, n, m, k);
59
60 // Natijani chop etish
61 printf("Natija matritsasi:\n");
62 printMatrix(C, n, k);
63
64 return 0;
65 }
PROBLEMS
TERMINAL
DEBUG CONSOLE
PORTS
COMMENTS
OUTPUT
C/C++ Compil
8
Natija matritsasi:
50 60
35 48
admin@Jahongir-2 output % cd "/Users/admin/Desktop/dead/OT/2/output"
admin@Jahongir-2 output % ./"main3"
Satrni kiriting: Hello world
So'zlar:
Hello
world
admin@Jahongir-2 output % cd "/Users/admin/Desktop/dead/OT/2/output"
admin@Jahongir-2 output % ./"main"
A matritsasining o'lchamlarini kiriting (n m): 2 2
B matritsasining o'lchamlarini kiriting (m k): 2 2
A matritsasini kiriting:
1 2 3 4
B matritsasini kiriting:
5 6 7 8
Natija matritsasi:
19 22
43 50
admin@Jahongir-2 output %


EXPLORER
kmlv.c
main.c
main2.c
main3.c
OPEN EDITORS
kmlv.c 1
main.c 2
main2.c 2
main3.c 2
OT
1
2
output
main.c
main2.c
main3.c
output
OUTLINE
TIMELINE
MYSQL
2 > C main2.c > main()
26 int main() {
49 // B matritsasini kiritish
50 printf("B matritsasini kiriting:\n");
51 for (int i = 0; i < m; i++) {
52 for (int j = 0; j < k; j++) {
53 scanf("%d", &B[i * k + j]);
54 }
55 }
56
57 // Matritsalarni ko'paytirish
58 multiplyMatrices1D(A, B, C, n, m, k);
59
60 // Natijani chop etish
61 printf("Natija matritsasi:\n");
62 printMatrix1D(C, n, k);
63
64 return 0;
65 }
PROBLEMS
TERMINAL
DEBUG CONSOLE
PORTS
COMMENTS
OUTPUT
admin@Jahongir-2 output % cd "/Users/admin/Desktop/dead/OT/2/output"
admin@Jahongir-2 output % ./"main2"
A matritsasining o'lchamlarini kiriting (n m): 2
2
B matritsasining o'lchamlarini kiriting (m k): 2 2
A matritsasini kiriting:
1 2 3 4
B matritsasini kiriting:
4 3 2 1
Natija matritsasi:
8 5
20 13
admin@Jahongir-2 output %
0 0 0
Compile & Run
Compile
Debug
Ln 27, Col 17
Spaces:

2.2

9	Kiritilgan satrni so'zlarga ajratib, har bir so'zni alohida ajratilgan xotirada char ** massivga saqlang.
10	Dinamik massivdan juft va tog sonlarni ikkita yangi, aniq o'lchamli dinamik



GitHub repository: https://github.com/komilov6/OT-deadline.git